

Topological Sorting

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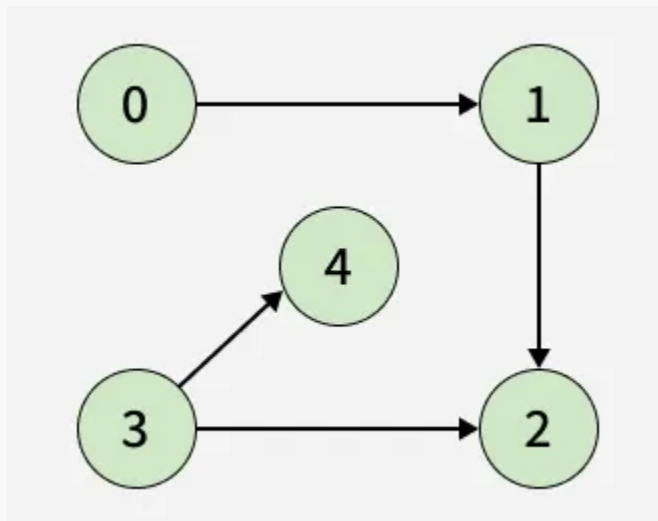
Topological sorting for Directed Acyclic Graph (DAG) is a linear ordering of vertices such that for every directed edge $u \rightarrow v$, vertex u comes before v in the ordering.

There may be several topological orderings for a graph.

Note: Topological Sorting for a graph is **not possible if the graph is not a DAG.**

Example:

Input: $adj[][] = [[1], [2], [], [2, 4], []]$



Output: $[0, 3, 1, 4, 2]$

Explanation: Since vertices 0 and 3 do not have any incoming edges from any other vertex, they appear first in the topological ordering. Next, vertex 1 depends only on vertex 0, so it comes after 0. Similarly, vertex 4 depends only on vertex 3, placing it after 3. Finally, vertex 2 depends on both vertices 1 and 3, so it appears after both of them in the ordering.

What is Topological Sort?

Topological Sort is **a linear ordering of vertices** in a Directed Acyclic Graph (DAG) such that for every directed edge $u \rightarrow v$, vertex u appears before v in the ordering.

- Applies only to **DAGs** (Directed Acyclic Graphs), and is not possible for **cyclic** or **undirected** graphs.
- It doesn't "traverse" all vertices like BFS/DFS; instead, it **represents dependency ordering between them.**

- Used in problems like **task scheduling, build order, course prerequisite ordering**, etc.

Topological Sorting in Directed Acyclic Graphs (DAGs)

DAGs are graphs where each edge is directed and no cycle exists in the graph. Before understanding why Topological sort only exists for DAGs, let's first answer two questions:

- **Why Topological Sort is not possible for graphs with undirected edges or cycles?**

Undirected Graph:

Every edge $u-v$ can be seen as both $u \rightarrow v$ and $v \rightarrow u$, so you **can't decide** which one should come first.

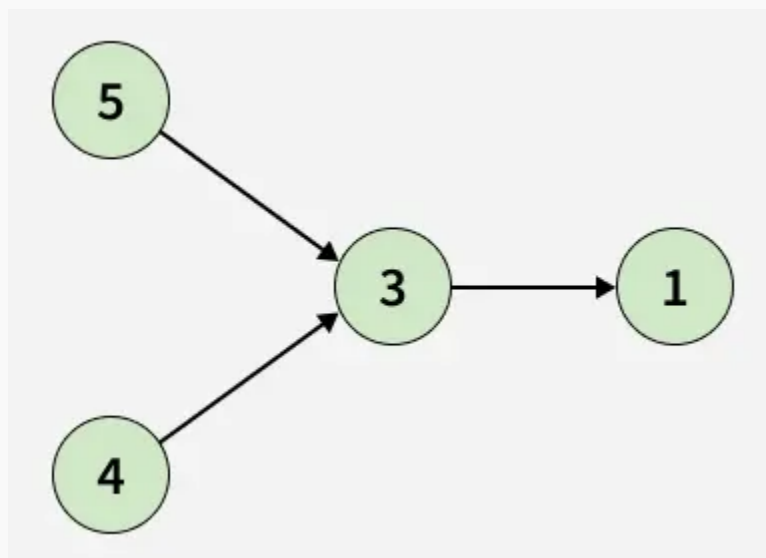
Cyclic Graph:

Let's say the graph contains the cycle, $u \rightarrow v \rightarrow w \rightarrow u$, then you'd need u before v , v before w , and w before u — **impossible to satisfy**.

- **Topological order may not be Unique:**

Topological sort represents all possible ordering satisfying the condition that if there is an edge between $u \rightarrow v$, u comes before v in the ordering. Any ordering that satisfies this for all edges ($u \rightarrow v$) is valid.

Let's see all possible topological orderings for the below graph:



For the above image, $4 \rightarrow 5 \rightarrow 3 \rightarrow 1$ and $5 \rightarrow 4 \rightarrow 3 \rightarrow 1$, both topological orders are valid.

Topological Sorting Using DFS:

The main idea is **to perform a Depth First Search (DFS)** on the Directed Acyclic Graph (DAG) and, for each vertex, push it onto a stack only after visiting all its adjacent vertices. This ensures that every vertex appears after all its neighboring vertices.

Finally, **reversing the stack (or popping elements from it)** gives the topological ordering of the graph.

Refer to [this article](#) to read more.

Topological Sorting Using BFS:

Topological Sort using BFS (**Kahn's Algorithm**) works by repeatedly selecting vertices with **in-degree zero (no dependencies)**, adding them to the result, and reducing the in-degree of their adjacent vertices. This process continues until all vertices are processed, producing a valid linear ordering of a DAG.

Refer to [this article](#) to read more.

Advantages of Topological Sort:

- Helps in scheduling tasks or events based on dependencies.
- **Detects cycles in a directed graph.**
- Efficient for solving problems with precedence constraints.

Disadvantages of Topological Sort:

- Only applicable to directed acyclic graphs (DAGs), not suitable for cyclic graphs.
- May not be unique, multiple valid topological orderings can exist.

Applications of Topological Sort:

- Task scheduling and project management.
- In software deployment tools like Makefile.

- Dependency resolution in package management systems.
- Determining the order of compilation in software build systems.
- Deadlock detection in operating systems.
- Course scheduling in universities.
- [Kahn's algorithm for Topological Sorting](#)
- [All Topological Sorts of a Directed Acyclic Graph](#)
- [Topological sort using DFS](#)

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